FINAL DRAFT — CLASS A PLAN (Base: 10.22.0.0/16)

Company: _____
Class A Base: 10.22.0.0/16
Site summaries (for routing summarization):

- Head Office (HO): **10.22.0.0/20**
- Mpumalanga (MPU): **10.22.16.0/20**
- Limpopo (LIM): **10.22.32.0/20**
- Eastern Cape (EC): **10.22.48.0/20**
- WAN/Loopbacks: **10.22.240.0/24** (use /30s inside)

A. Head Office (Gauteng) — VLANs, Subnets, Gateways, DHCP, Ports

VLAN Name	VLAN ID	Subnet (mask)	Default GW (SVI / Router Subif)	DHCP Range (router HO)	Switch ports (SW-HO-CORE)	Router subinterface
IT	10	10.22.0.0 /27 (30 hosts)	**10.22.0.1**	10.22.0.2 – 10.22.0.30	Fa0/1 – Fa0/5	R-HO G0/1.10 (encap 802.1Q 10)
HR	20	10.22.0.32 /27	**10.22.0.33**	10.22.0.34 – 10.22.0.62	Fa0/6 – Fa0/10	R-HO G0/1.20
Procurement	30	10.22.0.64 /27	**10.22.0.65**	10.22.0.66 – 10.22.0.94	Fa0/11 – Fa0/15	R-HO G0/1.30
Finance	40	10.22.0.96 /27	**10.22.0.97**	10.22.0.98 – 10.22.0.126	Fa0/16 – Fa0/20	R-HO G0/1.40
Server Farm (Servers VLAN)	50	10.22.0.128 /26 (62 hosts)	**10.22.0.129**	(static servers; small DHCP if needed 10.22.0.130–149)	Fa0/21 – Fa0/23 (servers)	R-HO G0/1.50
Management / Shared Devices	99	10.22.0.192 /27	**10.22.0.193**	10.22.0.194 – 10.22.0.222	Fa0/24 + spare	R-HO G0/1.99

Server static IP suggestions (in Server VLAN):

- DNS Server: **10.22.0.130** (set in DHCP as DNS server)
- File Server: **10.22.0.131**

- Web Server: **10.22.0.132**
- (If you later run DHCP on server) DHCP Server: **10.22.0.133** — *but we are using router DHCP for HO scopes*

Notes (HO):

- Router R-HO handles DHCP pools for HO VLANs (router-on-a-stick).
- SW-HO-CORE Gi0/1 is trunk to R-HO G0/1 (allow VLANs 10,20,30,40,50,99).
- Use Fa0/1–Fa0/20 for the 20 department PCs and Fa0/21–Fa0/23 for servers. Fa0/24 reserved for expansion/management.

B. Mpumalanga Branch — VLANs, Subnets, Gateways, DHCP, Ports (includes Voice VLAN)

VLAN Name	VLAN ID	Subnet	Default GW	DHCP Range (R-MPU)	Switch ports (SW-MPU-1)	Router subinterface
Transport	110	10.22.16.0 /27	**10.22.16.1**	10.22.16.2 – 10.22.16.30	Fa0/1 – Fa0/5	R-MPU G0/1.110
Student Centre	120	10.22.16.32 /27	**10.22.16.33**	10.22.16.34 – 10.22.16.62	Fa0/6 – Fa0/10	R-MPU G0/1.120
Voice (IP Phones)	150	10.22.16.64 /26 (62 hosts)	**10.22.16.65**	10.22.16.66 – 10.22.16.126	same desk ports (phone passthrough) Fa0/1–Fa0/10	R-MPU G0/1.150
Mgmt / Printers	199	10.22.16.128 /27	**10.22.16.129**	10.22.16.130 – 10.22.16.158	Fa0/21 (printer), Fa0/22 (admin)	R-MPU G0/1.199

Ports & Voice notes (MPU):

- Use PoE on SW-MPU if simulating phones; configure each desktop port with switchport access vlan {110|120} and switchport voice vlan 150.
- If phone passthrough used, each desk uses one port (phone in, PC connected to phone).
- R-MPU G0/0 = WAN to HO: **10.22.240.2/30** (HO .1)

C. Limpopo Branch — VLANs, Subnets, Gateways, DHCP, Ports

VLAN Name	VLAN ID	Subnet	Default GW	DHCP Range (R-LIM)	Switch ports (SW-LIM)	Router subinterface
Users	210	10.22.32.0 /27	**10.22.32.1**	10.22.32.2 – 10.22.32.30	Fa0/1 – Fa0/10	R-LIM G0/1.210

CONFIDENTIAL

Mgmt / Printers	299	10.22.32.32 /27	**10.22.32.33**	10.22.32.34 – 10.22.32.62	Fa0/11 – Fa0/12	R-LIM G0/1.299

Ports (LIM):

- Use Fa0/1–Fa0/10 for the 10 user PCs.
- R-LIM G0/0 (WAN) = **10.22.240.6/30** (HO side .5)

D. Eastern Cape Branch — VLANs, Subnets, Gateways, DHCP, Ports

VLAN Name	VLAN ID	Subnet	Default GW	DHCP Range (R-EC)	Switch ports (SW-EC)	Router subinterface
Users	310	10.22.48.0 /27	**10.22.48.1**	10.22.48.2 – 10.22.48.30	Fa0/1 – Fa0/10	R-EC G0/1.310
Mgmt / Printers	399	10.22.48.32 /27	**10.22.48.33**	10.22.48.34 – 10.22.48.62	Fa0/11 – Fa0/12	R-EC G0/1.399

Ports (EC):

- Fa0/1–Fa0/10 for PCs, Fa0/11 printer, Fa0/12 admin.
- R-EC G0/0 (WAN) = **10.22.240.10/30** (HO side .9)

E. WAN / Inter-site Links (Class A addresses, /30s)

Link	Subnet (/30)	HO Router IP	Branch Router IP
HO ↔ Mpumalanga	10.22.240.0 /30	**10.22.240.1**	**10.22.240.2**
HO ↔ Limpopo	10.22.240.4 /30	**10.22.240.5**	**10.22.240.6**
HO ↔ Eastern Cape	10.22.240.8 /30	**10.22.240.9**	**10.22.240.10**
HO ↔ ISP	10.22.240.12 /30	**10.22.240.13**	ISP **10.22.240.14**

Router physical/interface summary:

- **R-HO**:
  - G0/0 ⟶ WAN (ISP) 10.22.240.13/30 (or separate physical links to branches if desired)
  - G0/1 ⟶ Trunk to SW-HO-CORE (subinterfaces G0/1.10, .20, .30, .40, .50, .99)
- **R-MPU**:
  - G0/0 ⟶ WAN to HO 10.22.240.2/30

CONFIDEN

- G0/1 ⟶ Trunk to SW-MPU-1 (subinterfaces .110, .120, .150, .199)

- **R-LIM**: G0/0 WAN 10.22.240.6/30; G0/1 trunk with subifs .210, .299
- **R-EC**: G0/0 WAN 10.22.240.10/30; G0/1 trunk with subifs .310, .399

F. DHCP Configuration Notes (router-based DHCP)

- **HO router** (R-HO) has DHCP pools for HO VLANs:
  - Example CLI (concept):
  - ip dhcp pool IT
  - network 10.22.0.0 255.255.255.224
  - default-router 10.22.0.1
  - dns-server 10.22.0.130
  - lease 2
- **Branch routers** (R-MPU / R-LIM / R-EC) host DHCP pools for their local VLANs similarly.
- **DHCP Exclusions:** exclude gateway & server & WAN interface addresses from scopes (e.g., ip dhcp excluded-address 10.22.0.1 10.22.0.5 as needed).

G. Server Farm Details (HQ)

Server	Role	Static IP (Server VLAN 50)	Notes
DNS Server	name resolution	10.22.0.130	Set as DNS in DHCP pools
File Server	shared storage	10.22.0.131	Set share and perms per dept
Web Server	intranet/web app	10.22.0.132	HR allowed to access (ACL rule)
(Optional) Backup/DB	backup or DB	10.22.0.133	as needed

Server VLAN SVI / Router subif: 10.22.0.129 (R-HO G0/1.50)

H. Access Port Mapping

Head Office SW-HO-CORE (Fa0/1 – Fa0/24):

- Fa0/1–Fa0/5 ⟶ VLAN 10 (IT PCs)
- Fa0/6–Fa0/10 ⟶ VLAN 20 (HR PCs)
- Fa0/11–Fa0/15 ⟶ VLAN 30 (Procurement PCs)

CONFIDENTIAL

- Fa0/16–Fa0/20 → VLAN 40 (Finance PCs)
- Fa0/21 → Server DNS (VLAN 50, static 10.22.0.130)
- Fa0/22 → File Server (VLAN 50, static 10.22.0.131)
- Fa0/23 → Web Server (VLAN 50, static 10.22.0.132)
- Fa0/24 → Management/Shared (VLAN 99)

MPU SW-MPU-1:

- Fa0/1–Fa0/5 → VLAN 110 (Transport; voice passthrough with voice VLAN 150)
- Fa0/6–Fa0/10 → VLAN 120 (Student Centre; voice passthrough)
- Fa0/21 → Printer (VLAN 199)
- Gi0/1 → Trunk to R-MPU G0/1 (allow 110,120,150,199)

LIM SW-LIM-1:

- Fa0/1–Fa0/10 → VLAN 210 (users)
- Fa0/11 → Printer (VLAN 299)
- Gi0/1 → Trunk to R-LIM G0/1

EC SW-EC-1:

- Fa0/1–Fa0/10 → VLAN 310 (users)
- Fa0/11 → Printer (VLAN 399)
- Gi0/1 → Trunk to R-EC G0/1

I. ACL & Security Notes (final draft reminders)

- **ACL Example rules (HQ):**
    - Permit HR VLAN (20) → Web Server (10.22.0.132) TCP 80/443
    - Deny other VLANs → Web Server (unless required)
    - Permit IT VLAN (10) → File Server (10.22.0.131) SMB (or general)
    - Allow DNS (UDP 53) from all VLANs to DNS Server (10.22.0.130)
- Apply ACLs on **router SVIs** or **near destination** (e.g., inbound to server SVI).
- Document each ACL rule in the final report and test from representative hosts.

CONFIDENTIAL

J. Testing Checklist (final)

- Trunk up: R-HO G0/1 ↔ SW-HO Gi0/1 (VLANs allowed)
- Each PC gets IP from correct DHCP scope and correct default gateway
- PCs can ping their gateway and HQ servers (DNS, File, Web) as permissions allow
- Branch PCs can reach HO servers and Internet (traceroute + ping)
- Voice VLAN phones obtain IPs in MPU and can register (if simulated)
- ACLs restrict/allow access per policy (verify with ping/TCP tests)
- Save & version master .pka in shared Drive; label versions clearly

CONFIDENTIAL